

# Supplementary Materials: Mapping the Antigenic Landscape of DENV and ZIKV via PLM Embeddings and Graph Topology

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**Abstract.** [Abstract of the Supplementary Materials] This document provides supplementary material for the article *Mapping the Antigenic Landscape of DENV and ZIKV via PLM Embeddings and Graph Topology*. The study integrates ProtT5 protein language model embeddings with K-Means clustering and graph-based similarity analysis to identify hyperconnected, conserved epitopes across Dengue (DENV) serotypes and Zika (ZIKV) lineages from a curated dataset of 5,820 unique epitopes. This supplement includes: (i) the formal definition of the multicriteria cluster scoring system ( $F_{dens}$ ,  $F_{eq}$ ,  $F_{zika}$ , Eqs. S1–S4); (ii) cluster-level summary tables (Tables S1–S12) for each of the four analytical groups; and (iii) general cluster maps (Figs. S1–S3) and the alignment distribution figure (Fig. S4) supporting the results of the main manuscript.

**Keywords:** Epitopes · Dengue · Zika · Immunoinformatics · Large Language Models · Graph Analysis.

## 1 Cluster Evaluation and Multicriteria Ranking Metrics

To ensure the reproducibility of the cluster evaluation process and systematically address the natural under-representation of ZIKV sequences in the raw dataset, a multi-criteria scoring system was formally defined. The algorithm calculates three primary metrics for each cluster:

**1. Normalized Density Factor ( $F_{dens}$ ):** Spatial compactness is determined by dividing the total number of sequences within a cluster by the area of its convex hull in the reduced dimensionality space. This raw density ( $D$ ) is then normalized using Min-Max scaling:

$$F_{dens} = \frac{D_i - D_{min}}{D_{max} - D_{min}} \quad (S1)$$

Where  $D = \infty$  is capped at  $F_{dens} = 1.0$ .

**2. Equilibrium Factor ( $F_{eq}$ ):** This metric penalizes extreme imbalances between the viral species, rewarding clusters where the proportions of DENV ( $P_{DENV}$ ) and ZIKV ( $P_{ZIKV}$ ) approach parity:

$$F_{eq} = 1 - \frac{|P_{DENV} - P_{ZIKV}|}{100} \quad (S2)$$

**3. Zika Representativeness Factor ( $F_{zika}$ ):** This factor captures the direct percentage of ZIKV sequences present in the cluster:

$$F_{zika} = \frac{P_{ZIKV}}{100} \quad (S3)$$

**Composite Final Score and Weighting Scheme:** To actively compensate for the dataset’s bias toward DENV and avoid systematically under-ranking ZIKV-enriched regions, the metrics are combined multiplicatively with an exponential weighting scheme. An elevated exponential weight of 2.5 is exclusively applied to the  $F_{zika}$  parameter. Furthermore, a penalty factor of 0.1 is applied if a cluster fails to meet a minimum threshold of 5.0% ZIKV sequences:

$$Score = F_{dens} \times F_{eq} \times F_{zika}^{2.5} \quad (S4)$$

## 2 Cluster Summary Tables

The following tables report cluster-level statistics for each of the four analytical groups. For each group, three tables are provided: (i) cluster description with compositional metrics; (ii) alignment characteristics per cluster; and (iii) a multicriteria ranking of the top three clusters, computed as described in Sect. 1. The cluster map for Group I is presented as Fig. 3 of the main manuscript; the maps for Groups II–IV are shown in Figs. S1–S3 of this supplement. IE: Immunogenic Epitopes; NIE: Non-Immunogenic Epitopes; EAPE: Epitopes Aligned to Protein E; RP: Reference Proteome.

### 2.1 Group I: Immunogenic and Non-Immunogenic Epitopes

**Table S1.** Cluster Description of Immunogenic and Non-Immunogenic Epitopes (5 clusters). All percentages are based exclusively on aligned epitopes; unaligned epitopes were not classified. Epitopes aligned with both viruses were counted separately. Column abbreviations — Cl.: cluster; %DENV: percentage of aligned epitopes matching Dengue; %ZIKV: percentage matching Zika; D&Z: epitopes aligned to both viruses; T.DENV/T.ZIKV: total DENV/ZIKV epitopes in the cluster; Total: all epitopes; Area: convex hull polygon area in the embedding space; Dens.: epitopes per unit area.

| Cl. | %DENV | %ZIKV | D&Z | T.DENV | T.ZIKV | Total | Area  | Dens. |
|-----|-------|-------|-----|--------|--------|-------|-------|-------|
| 0   | 51.37 | 19.42 | 6   | 582    | 220    | 1133  | 49.33 | 22.97 |
| 1   | 54.04 | 18.42 | 8   | 789    | 269    | 1460  | 30.97 | 47.14 |
| 2   | 55.38 | 20.07 | 9   | 679    | 246    | 1226  | 77.99 | 15.77 |
| 3   | 78.31 | 8.92  | 36  | 527    | 60     | 673   | 37.10 | 18.14 |
| 4   | 51.37 | 19.42 | 3   | 640    | 252    | 1302  | 43.93 | 29.64 |

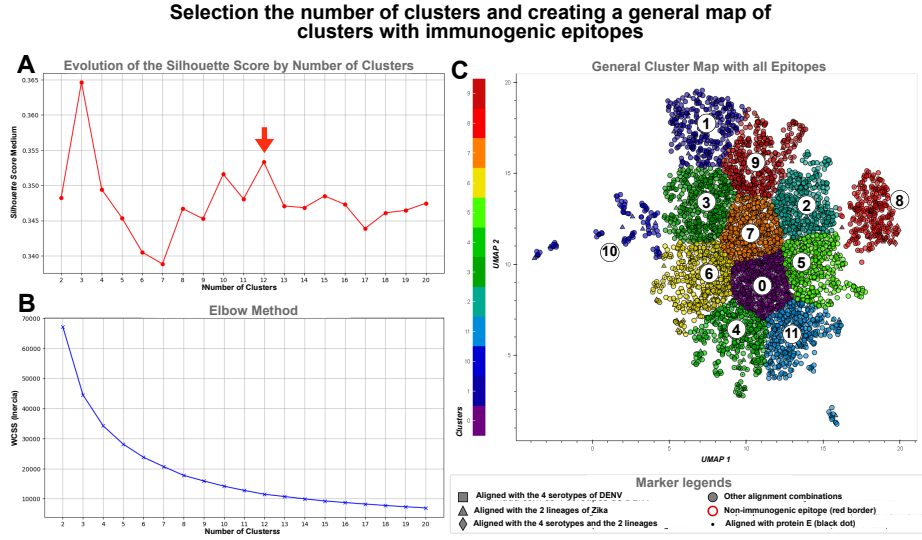
**Table S2.** Alignment Characteristics per Cluster of Immunogenic and Non-Immunogenic Epitopes. Column abbreviations — Cl.: cluster; PE-Al.: epitopes aligned to Protein E; IE: immunogenic epitopes; NIE: non-immunogenic epitopes; All RPs: epitopes matching all six reference proteomes (DENV serotypes 1–4, ZIKV African and Asian lineages).

| Cl. | PE-Al. | IE   | NIE | All RPs |
|-----|--------|------|-----|---------|
| 0   | 136    | 997  | 136 | 0       |
| 1   | 161    | 1189 | 271 | 1       |
| 2   | 223    | 990  | 236 | 1       |
| 3   | 427    | 373  | 300 | 18      |
| 4   | 237    | 1079 | 223 | 0       |

**Table S3.** Cluster Ranking of Immunogenic and Non-Immunogenic Epitopes. Only the top three clusters are shown. Metrics computed as defined in Sect. 1. Column abbreviations — Cl.: cluster;  $F_{dens}$ : normalized density factor;  $F_{eq}$ : equilibrium factor;  $F_{zika}$ : Zika representativeness factor; Score: composite final score ( $F_{dens} \times F_{eq} \times F_{zika}^{2.5}$ ).

| Rank            | Cl. | $F_{dens}$ | $F_{eq}$ | $F_{zika}$ | Score  |
|-----------------|-----|------------|----------|------------|--------|
| 1 <sup>st</sup> | 1   | 1.00       | 0.64     | 0.18       | 0.0094 |
| 2 <sup>nd</sup> | 4   | 0.44       | 0.70     | 0.19       | 0.0051 |
| 3 <sup>rd</sup> | 0   | 0.23       | 0.68     | 0.19       | 0.0026 |

## 2.2 Group II: Immunogenic Epitopes Only



**Fig. S1.** Selection of the number of Clusters and General Cluster Map with Immunogenic Epitopes. **A** – Evolution of Silhouette Score by number of Clusters, graph demonstrating the scores tested according to divisions from 2 to 20, highlighting the division into 12 Clusters (Red Arrow), score of 0.353 (Highest score above 3 divisions). **B** – Elbow Method, demonstrating the Sum or Inertia of Squares Within Clusters (WCSS - Within-Cluster Sum of Squares) by the number of Clusters grouped by K-Means, highlighting the points of least inflection from 5 to 7 Clusters. **C** – General Cluster Map only with the collected Immunogenic Epitopes, demonstrating the division into 12 regions, of different colors, named from 0 to 11, with epitopes plotted in the most varied forms in the embedding space (Umap2, Umap1).

**Table S4.** Cluster Description of Immunogenic Epitopes (12 clusters). All percentages are based exclusively on aligned epitopes. Epitopes aligned with both viruses were counted separately. Column abbreviations — Cl.: cluster; %DENV: percentage of aligned epitopes matching Dengue; %ZIKV: percentage matching Zika; D&Z: epitopes aligned to both viruses; T.DENV/T.ZIKV: total DENV/ZIKV epitopes; Total: all epitopes; Area: convex hull polygon area; Dens.: epitopes per unit area.

| Cl. | %DENV | %ZIKV | D&Z | T.DENV | T.ZIKV | Total | Area  | Dens. |
|-----|-------|-------|-----|--------|--------|-------|-------|-------|
| 0   | 51.33 | 19.54 | 3   | 289    | 110    | 563   | 10.32 | 54.55 |
| 1   | 46.88 | 17.01 | 0   | 135    | 49     | 288   | 15.96 | 18.04 |
| 2   | 50.64 | 17.99 | 2   | 197    | 70     | 389   | 14.52 | 26.79 |
| 3   | 52.11 | 17.52 | 4   | 235    | 79     | 451   | 15.30 | 29.47 |
| 4   | 52.97 | 17.84 | 0   | 196    | 66     | 370   | 17.60 | 21.02 |
| 5   | 51.56 | 16.52 | 0   | 231    | 74     | 448   | 15.58 | 28.75 |
| 6   | 57.24 | 21.27 | 1   | 261    | 97     | 456   | 19.01 | 23.99 |
| 7   | 54.88 | 19.69 | 5   | 315    | 113    | 574   | 11.17 | 51.39 |
| 8   | 86.75 | 11.54 | 27  | 203    | 27     | 234   | 12.96 | 18.06 |
| 9   | 52.25 | 21.49 | 3   | 197    | 81     | 377   | 17.11 | 22.04 |
| 10  | 61.70 | 20.21 | 3   | 58     | 19     | 94    | 18.09 | 5.20  |
| 11  | 47.66 | 16.67 | 0   | 183    | 64     | 384   | 23.89 | 16.08 |

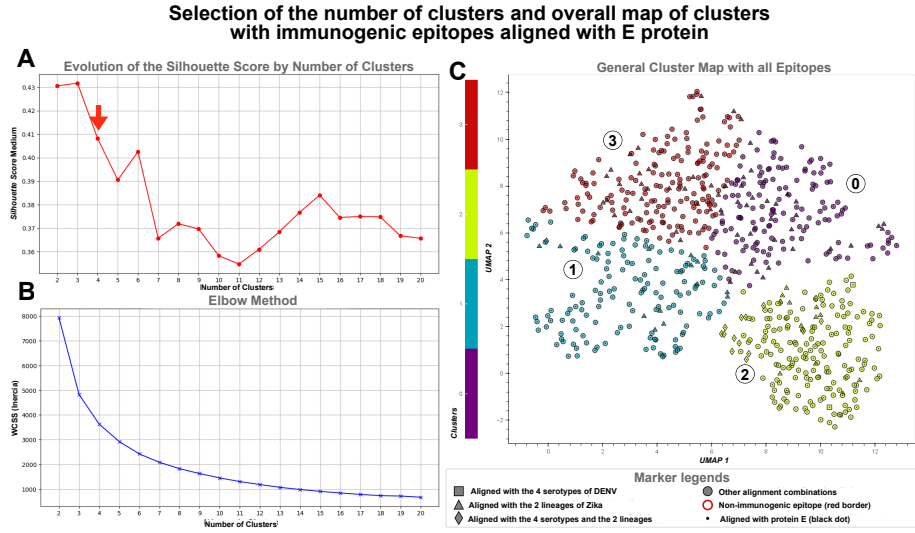
**Table S5.** Alignment Characteristics per Cluster of Immunogenic Epitopes. Non-immunogenic epitopes were excluded from this group (NIE = 0 for all clusters). Column abbreviations — Cl.: cluster; PE-Al.: epitopes aligned to Protein E; IE: immunogenic epitopes; NIE: non-immunogenic epitopes; All RPs: epitopes matching all six reference proteomes.

| Cl. | PE-Al. | IE  | NIE | All RPs |
|-----|--------|-----|-----|---------|
| 0   | 83     | 563 | 0   | 0       |
| 1   | 30     | 288 | 0   | 0       |
| 2   | 21     | 389 | 0   | 0       |
| 3   | 45     | 451 | 0   | 1       |
| 4   | 85     | 370 | 0   | 0       |
| 5   | 9      | 448 | 0   | 0       |
| 6   | 93     | 456 | 0   | 1       |
| 7   | 84     | 574 | 0   | 0       |
| 8   | 195    | 234 | 0   | 15      |
| 9   | 40     | 377 | 0   | 0       |
| 10  | 1      | 94  | 0   | 0       |
| 11  | 35     | 384 | 0   | 0       |

**Table S6.** Cluster Ranking of Immunogenic Epitopes. Only the top three clusters are shown. Column abbreviations — Cl.: cluster;  $F_{dens}$ : normalized density factor;  $F_{eq}$ : equilibrium factor;  $F_{zika}$ : Zika representativeness factor; Score: composite final score ( $F_{dens} \times F_{eq} \times F_{zika}^{2.5}$ ).

| Rank            | Cl. | $F_{dens}$ | $F_{eq}$ | $F_{zika}$ | Score  |
|-----------------|-----|------------|----------|------------|--------|
| 1 <sup>st</sup> | 0   | 1.00       | 0.68     | 0.195      | 0.0115 |
| 2 <sup>nd</sup> | 7   | 0.94       | 0.65     | 0.197      | 0.0104 |
| 3 <sup>rd</sup> | 6   | 0.38       | 0.64     | 0.213      | 0.0051 |

### 2.3 Group III: Immunogenic and Non-Immunogenic Epitopes Aligned to Protein E



**Fig. S2.** Selection of the number of Clusters and General Cluster Map with Immunogenic and non-Immunogenic Epitopes Aligned with Protein E. **A** – Evolution of Silhouette Score by number of Clusters, graph showing the scores tested according to divisions from 2 to 20, highlighting the division into 4 Clusters (Red Arrow), score of 0.444 (Highest score above 3 divisions). **B** – Elbow Method, showing the Sum or Inertia of Squares Within Clusters (WCSS - Within-Cluster Sum of Squares) by the number of Clusters grouped by K-Means, highlighting the points of least inflection from 4 to 6 Clusters. **C** – General Cluster Map with only immunogenic and non-immunogenic epitopes aligned to PE, demonstrating the division into 4 regions, of different colors, named from 0 to 3, with epitopes plotted in the most different forms in the embedding space (Umap2, Umap1).

**Table S7.** Cluster Description of Immunogenic and Non-Immunogenic Epitopes Aligned to Protein E (4 clusters). All percentages are based on aligned epitopes only. Epitopes aligned with both viruses were counted separately. Column abbreviations — Cl.: cluster; %DENV: percentage of aligned epitopes matching Dengue; %ZIKV: percentage matching Zika; D&Z: epitopes aligned to both viruses; T.DENV/T.ZIKV: total DENV/ZIKV epitopes; Total: all epitopes; Area: convex hull polygon area; Dens.: epitopes per unit area.

| Cl. | %DENV | %ZIKV | D&Z | T.DENV | T.ZIKV | Total | Area  | Dens. |
|-----|-------|-------|-----|--------|--------|-------|-------|-------|
| 0   | 71.92 | 28.08 | 0   | 146    | 57     | 203   | 35.88 | 5.66  |
| 1   | 76.74 | 24.03 | 2   | 198    | 62     | 258   | 34.74 | 7.43  |
| 2   | 99.77 | 8.69  | 36  | 425    | 37     | 426   | 39.33 | 10.83 |
| 3   | 73.06 | 26.94 | 0   | 217    | 80     | 297   | 38.90 | 7.64  |

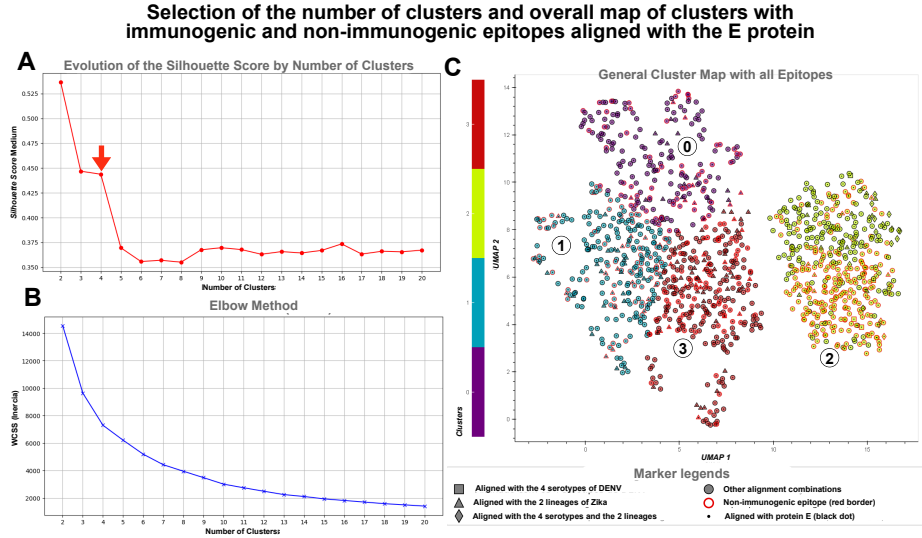
**Table S8.** Alignment Characteristics per Cluster of Immunogenic and Non-Immunogenic Epitopes Aligned to Protein E. All epitopes in this group are Protein E-aligned (PE-Al. = Total). Column abbreviations — Cl.: cluster; PE-Al.: epitopes aligned to Protein E; IE: immunogenic epitopes; NIE: non-immunogenic epitopes; All RPs: epitopes matching all six reference proteomes.

| Cl. | PE-Al. | IE  | NIE | All RPs |
|-----|--------|-----|-----|---------|
| 0   | 203    | 137 | 66  | 0       |
| 1   | 258    | 178 | 80  | 1       |
| 2   | 426    | 197 | 229 | 18      |
| 3   | 297    | 209 | 88  | 0       |

**Table S9.** Cluster Ranking of Immunogenic and Non-Immunogenic Epitopes Aligned to Protein E. Only the top three clusters are shown. Column abbreviations — Cl.: cluster;  $F_{dens}$ : normalized density factor;  $F_{eq}$ : equilibrium factor;  $F_{zika}$ : Zika representativeness factor; Score: composite final score ( $F_{dens} \times F_{eq} \times F_{zika}^{2.5}$ ).

| Rank            | Cl. | $F_{dens}$ | $F_{eq}$ | $F_{zika}$ | Score  |
|-----------------|-----|------------|----------|------------|--------|
| 1 <sup>st</sup> | 3   | 0.38       | 0.54     | 0.269      | 0.0078 |
| 2 <sup>nd</sup> | 1   | 0.34       | 0.47     | 0.240      | 0.0046 |
| 3 <sup>rd</sup> | 2   | 1.00       | 0.09     | 0.087      | 0.0002 |

## 2.4 Group IV: Immunogenic Epitopes Aligned to Protein E



**Fig. S3.** Selection of the number of Clusters and General Cluster Map with Immunogenic Epitopes Aligned with Protein E. **A** – Evolution of Silhouette Score by number of Clusters, graph demonstrating the tested scores according to divisions from 2 to 20, highlighting the division into 4 Clusters (Red Arrow), score of 0.408 (Highest score above 3 divisions). **B** – Elbow Method, demonstrating the Sum or Inertia of Squares Within Clusters (WCSS - Within-Cluster Sum of Squares) by the number of Clusters grouped by K-Means, highlighting the points of least inflection of 4 to 5 Clusters. **C** – General Map of Clusters with only the immunogenic epitopes aligned to PE, demonstrating the division into 4 regions, of different colors, named from 0 to 3, with epitopes plotted in the most varied forms in the embedding space (Umap2, Umap1).

**Table S10.** Cluster Description of Immunogenic Epitopes Aligned to Protein E (4 clusters). All percentages are based on aligned epitopes only. Epitopes aligned with both viruses were counted separately. Column abbreviations — Cl.: cluster; %DENV: percentage of aligned epitopes matching Dengue; %ZIKV: percentage matching Zika; D&Z: epitopes aligned to both viruses; T.DENV/T.ZIKV: total DENV/ZIKV epitopes; Total: all epitopes; Area: convex hull polygon area; Dens.: epitopes per unit area.

| Cl. | %DENV  | %ZIKV | D&Z | T.DENV | T.ZIKV | Total | Area  | Dens. |
|-----|--------|-------|-----|--------|--------|-------|-------|-------|
| 0   | 75.00  | 25.00 | 0   | 129    | 43     | 172   | 27.72 | 6.20  |
| 1   | 78.57  | 22.08 | 1   | 121    | 34     | 154   | 32.62 | 4.72  |
| 2   | 100.00 | 13.71 | 27  | 197    | 27     | 197   | 26.75 | 7.36  |
| 3   | 73.74  | 26.26 | 0   | 146    | 52     | 198   | 27.57 | 7.18  |



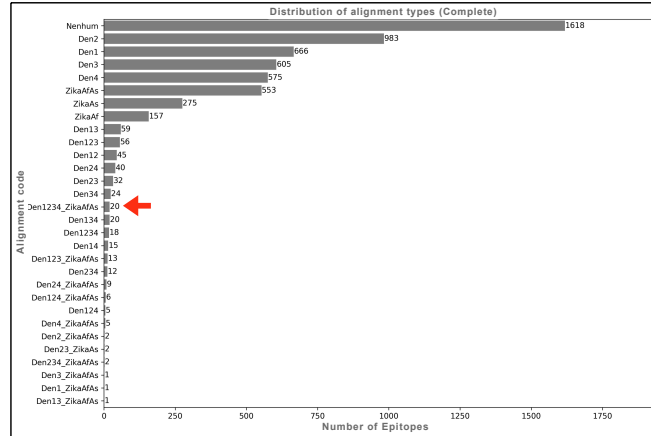
**Table S11.** Alignment Characteristics per Cluster of Immunogenic Epitopes Aligned to Protein E. Non-immunogenic epitopes were excluded ( $NIE = 0$ ); all epitopes are Protein E-aligned ( $PE-Al. = IE = Total$ ). Column abbreviations — Cl.: cluster; PE-Al.: epitopes aligned to Protein E; IE: immunogenic epitopes; NIE: non-immunogenic epitopes; All RPs: epitopes matching all six reference proteomes.

| Cl. | PE-Al. | IE  | NIE | All RPs |
|-----|--------|-----|-----|---------|
| 0   | 172    | 172 | 0   | 0       |
| 1   | 154    | 154 | 0   | 1       |
| 2   | 197    | 197 | 0   | 15      |
| 3   | 198    | 198 | 0   | 0       |

**Table S12.** Cluster Ranking of Immunogenic Epitopes Aligned to Protein E. Only the top three clusters are shown.  $F_{dens}$  values are normalized within this group’s density range ( $\max = 7.36$ ,  $\min = 4.72$ ). Column abbreviations — Cl.: cluster;  $F_{dens}$ : normalized density factor;  $F_{eq}$ : equilibrium factor;  $F_{zika}$ : Zika representativeness factor; Score: composite final score ( $F_{dens} \times F_{eq} \times F_{zika}^{2.5}$ ).

| Rank            | Cl. | $F_{dens}$ | $F_{eq}$ | $F_{zika}$ | Score  |
|-----------------|-----|------------|----------|------------|--------|
| 1 <sup>st</sup> | 3   | 0.93       | 0.53     | 0.263      | 0.0173 |
| 2 <sup>nd</sup> | 0   | 0.56       | 0.50     | 0.250      | 0.0088 |
| 3 <sup>rd</sup> | 2   | 1.00       | 0.14     | 0.137      | 0.0010 |

### 3 Alignment Distribution



**Fig. S4.** Distribution of Alignment Types. Alignment to Reference Proteomes (RPs) of all curated epitopes. Graph representing the number of epitopes (x-axis) per alignment that occurred (y-axis). Epitopes without alignment with RPs were classified as “No” alignment. Epitopes aligned with one or more RPs were counted under a code. On the y-axis are the alignment codes: The presence of “Den” indicates alignment with Dengue, the numbers 1 to 4 define the aligned serotypes; presence of “Zika” indicates alignment with Zika in the African lineage, represented by “Af”, and Asian, “As”; Epitopes that demonstrate alignment with different RPs accumulate codes.

**GitHub repository:** <https://github.com/LabMolUFG/Mapping-Antigenic-Landscape-via-PLM-and-Graph>